



NOMAD+

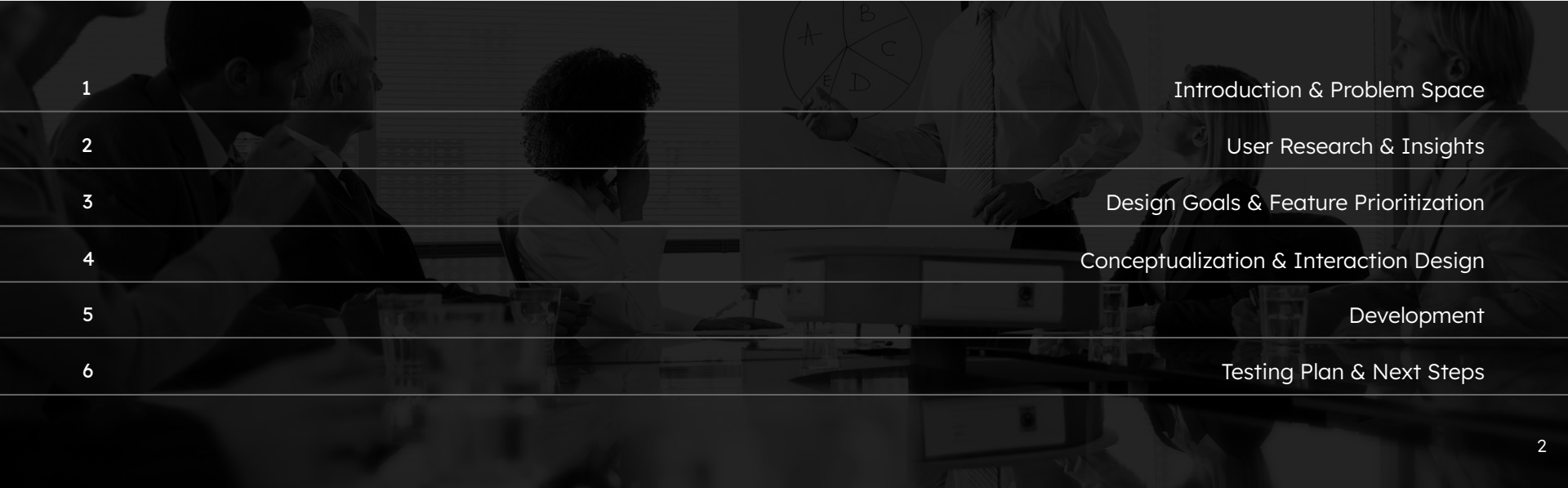
A Multimodal, Hands-Free Smart Kitchen Assistant

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INFO 882 - Fall 2025

AGENDA

Why we're here today

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- 1 Introduction & Problem Space
 - 2 User Research & Insights
 - 3 Design Goals & Feature Prioritization
 - 4 Conceptualization & Interaction Design
 - 5 Development
 - 6 Testing Plan & Next Steps

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INTRODUCTION & PROBLEM SPACE

PROJECT OVERVIEW

NOMAD+ is a multimodal smart kitchen assistant that helps home cooks navigate recipes without touching their devices.

- Combines voice input, gesture recognition, and future AR overlays to reduce interruptions while cooking.
- Helps users avoid touching screens with messy hands or constantly switching between apps.
- Keeps cooks focused on the food rather than device management.
- Explores how ubiquitous computing and multimodal interaction can enhance an everyday activity.
- Aims to create a more intuitive, accessible, and seamless cooking experience.

MOTIVATION

- I built the original concept of NOMAD+ in a previous class and wanted to advance it into a real, functional solution.
- I wanted to explore new interaction paradigms that go beyond traditional screens.
- Cooking is a natural use case for hands-free, multimodal interaction due to constant movement and messy hands.
- Existing tools do not fully support the realities of cooking, creating an opportunity for innovation.
- Developing the system allows me to directly address the problem space I identified through earlier research.
- This project aligns with my long-term goal of pursuing research in ubiquitous computing.

THE PROBLEM

- Cooking is messy, fast paced, and requires constant multitasking.
- Most recipe apps rely on touch-based interaction, which is difficult with wet or dirty hands.
- Users frequently lose their place in recipes due to scrolling, screen timeouts, or switching apps.
- Phones and tablets take up counter space and can get in the way while preparing food.
- Voice-only solutions are often too slow, inaccurate, or limited for real cooking workflows.
- These interruptions break the flow of cooking and add unnecessary friction to a routine task.

LIMITATIONS OF EXISTING PRODUCTS

- Most recipe apps are designed for touchscreens and do not account for messy or occupied hands.
- Scrolling, tapping, and navigating between multiple apps disrupts cooking flow.
- Voice assistants like Siri, Alexa, and Google Assistant offer limited recipe support and struggle with step-by-step pacing.
- Voice-only interfaces often misinterpret commands or provide incomplete information.
- Phones and tablets take up counter space and are prone to getting splashed, smudged, or knocked over.
- Visual learners lack spatial cues or demonstrations in traditional recipe formats.
- Existing tools do not combine multiple interaction modes in a cohesive way that matches real cooking behavior.

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USER RESEARCH & INSIGHTS

RESEARCH APPROACH

- Conducted a mixed-methods research process to understand real cooking behavior.
- Surveyed **60+ home cooks** to gather broad patterns and pain points.
- Followed up with **in-depth interviews** to explore behaviors, frustrations, and expectations in detail.
- Focused on how people use technology during cooking and where interruptions occur.
- Participants ranged from daily cooks to occasional cooks to capture diverse experiences.

RESEARCH QUESTIONS

- How do home cooks currently use technology while preparing meals?
- What moments of friction or interruption occur during the cooking process?
- How do users feel about voice, gesture, and AR as potential interaction modes?
- What types of hands-free interactions feel natural, intuitive, or distracting?
- What do users need from a tool to feel confident and supported while cooking?

KEY FINDINGS

- Phones are used constantly for recipes, timers, substitutions, and videos.
- Messy hands make touch screens inconvenient and frustrating.
- Screen timeouts and scrolling disrupt the cooking flow.
- Voice alone is not enough for fast-paced or complex steps.
- Gesture interaction feels natural and appealing for quick navigation.
- Visual learners want demonstrations or spatial cues that standard apps do not provide.
- Users strongly prefer tools that are simple, reliable, and low effort to set up.

INSIGHT SUMMARY

- Technology is essential in modern cooking, but current tools introduce friction.
- A successful solution must be multimodal, combining gesture, voice, and visual cues.
- Reducing cognitive load is critical; users should not have to think about the interface.
- There is a clear opportunity for a hands-free, context-aware recipe experience.
- These insights directly shaped the design direction for NOMAD+.

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DESIGN GOALS & FEATURE PRIORITIZATION

DESIGN GOALS

- Enable true hands-free cooking through gesture and voice interaction.
- Provide clear, segmented instructions that support visual learners.
- Reduce cognitive load by showing only what users need at each step.
- Create a system that feels simple, reliable, and trustworthy.
- Adapt to different cooking styles and experience levels.
- Ensure consistency across screen-based, gesture-based, and AR interactions.

MVP FEATURE PRIORITIES

High-Priority Features for MVP

- Simulated voice command interactions
 - Implemented as the core modality for the MVP
 - Chosen due to time constraints and increased development complexity
 - Allows users to step through a recipe and request basic information
- Clear step-by-step visual instructions
 - Minimal, readable UI to support recipe navigation
 - Reduces cognitive load and helps users stay oriented
- Audio feedback
 - Confirms when a command is processed
 - Reinforces user confidence during hands-free interaction

MVP FEATURE PRIORITIES (cont'd)

Deferred for Later Iterations

- **Gesture-based navigation**
 - Initially part of the core vision
 - Paused due to technical complexity and limited development time
- **AR overlays**
 - Early concepts planned
 - Will be explored once the main voice-based workflow is stable
- **Why This Pivot Makes Sense**
 - Allows the project to remain feasible within the quarter timeline
 - Preserves the integrity of the concept while narrowing scope
 - Ensures a functional, testable prototype that demonstrates the core value
 - Keeps gesture and AR pathways open for future research and development

WHY THESE FEATURES FIRST

- **Focuses on feasibility:**
 - Simulated voice commands provide a reliable way to demonstrate hands-free interaction within the limited development timeline.
- **Captures the core value of NOMAD+:**
 - Even without gestures and AR, voice interaction clearly conveys the vision of reducing cooking friction.
- **Reduces engineering complexity:**
 - Gesture recognition and AR overlays require additional modeling, sensor calibration, and performance tuning that exceed the quarter time frame.
- **Allows for a functional, testable prototype:**
 - A stable voice-based workflow offers meaningful user testing opportunities without relying on partially implemented modalities.
- **Sets the foundation for future multimodal expansion:**
 - Gesture interactions and AR overlays can be layered onto the existing architecture later as time and resources allow.
- **Keeps the project aligned with research insights:**
 - While gestures were highly desired by users, voice remains an essential and familiar mode of hands-free interaction.

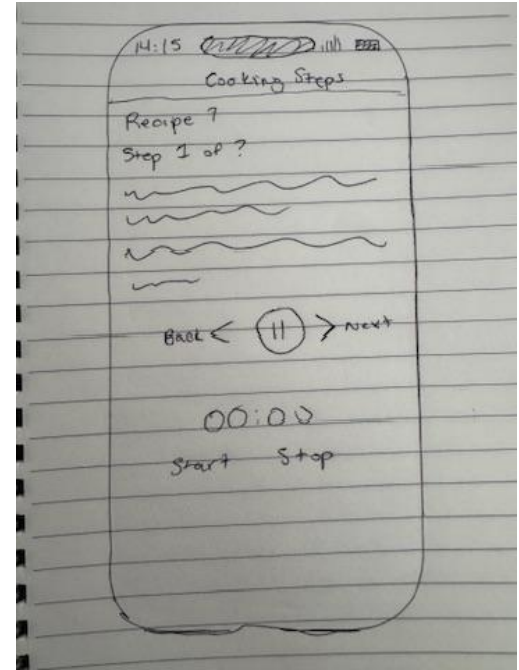
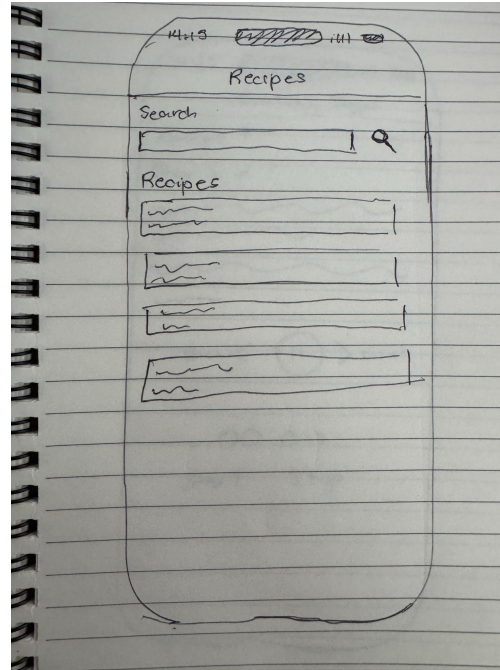
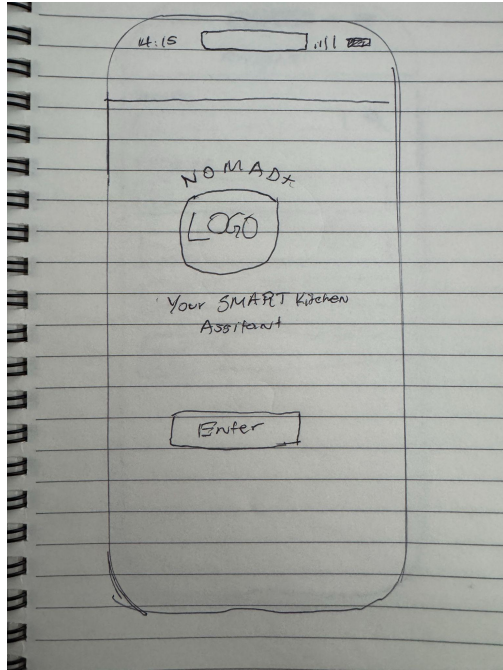
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CONCEPTUALIZATION & INTERACTION DESIGN

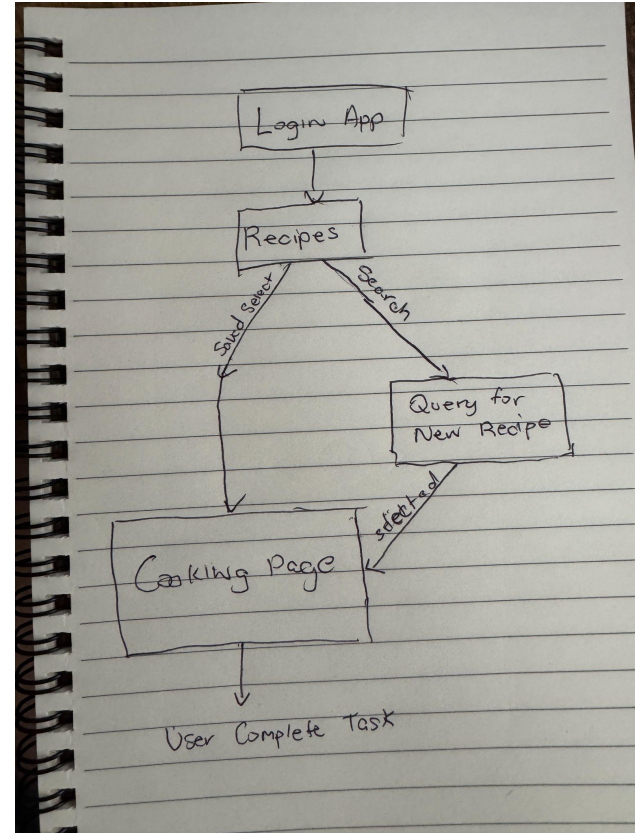
EARLY SKETCHES

- Explored foundational interaction flows for hands-free recipe navigation.
- Focused on reducing screen dependence and minimizing cognitive load.
- Sketched concepts for step-based navigation, timers, clarifications, and pacing.
- Evaluated different layout structures to support quick glances and comprehension.
- Used sketches to identify what information must be visible at all times vs. on demand.

INITIAL SKETCHES



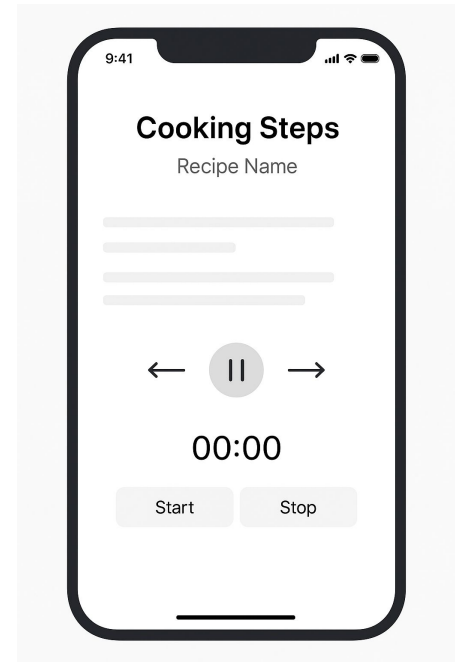
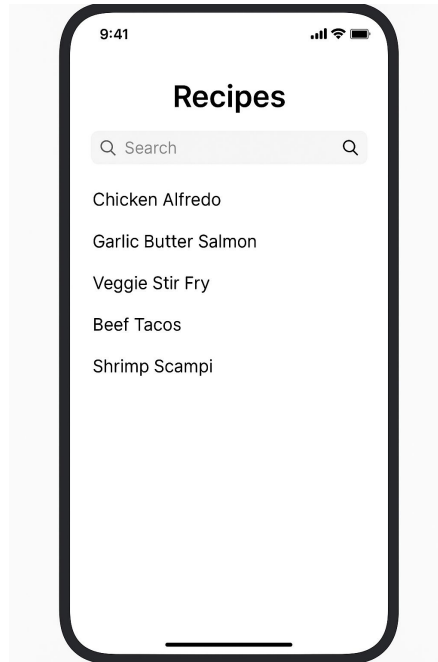
USER FLOW DIAGRAM



MID-FIDELITY WIREFRAMES

- Developed wireframes that emphasized clarity and simplicity.
- Introduced clean typography, segmented steps, and minimal visual clutter.
- Designed a consistent layout for step instructions, timing cues, and prompts.
- Ensured UI could function smoothly in a voice-only MVP.
- Prepared structure for future expansion into gesture and AR modes.

MID-FIDELITY WIREFRAMES (cont'd)



VOICE INTERACTION MODEL

- Voice serves as the primary hands-free input method in the MVP.
- Simulated commands allow users to:
 - Move to the next or previous step
 - Repeat steps
 - Request clarifications
 - Ask basic ingredient or timing questions
- Designed conversational prompts that feel natural and require minimal learning.
- Integrated audio confirmation cues so users know when commands are recognized.
- Forms the foundation for layering additional modalities later.

GESTURE INTERACTION CONCEPTS

- Originally planned as a core interaction method based on user enthusiasm.
- Gesture vocabulary included actions like swiping forward or pausing with a closed hand.
- Intent was to support intuitive, low-effort commands during cooking.
- Designs remain part of the long-term vision but are not implemented in the MVP due to complexity.
- Early conceptual work ensures gestures can be integrated smoothly later.

AR OVERLAY CONCEPTS

- Explored how AR could display step instructions in the user's environment.
- Early concepts include floating step cards, ingredient callouts, and simple spatial cues.
- Intended to support visual learners and reduce device dependence.
- AR will be implemented in later stages once the core workflow is fully validated.
- Planning now ensures consistent interaction patterns across modalities.

DESIGN PRINCIPLES GUIDING INTERACTION

- Low friction: Interfaces should require minimal mental effort.
- Predictability: Users always know what the system heard or understood.
- Clarity: Only essential information should appear at a time.
- Scalability: Architecture should support adding gesture and AR later without redesign.
- Multimodal cohesion: Voice, gestures, and AR should work together rather than feel separate.

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DEVELOPMENT

ARCHITECTURE OVERVIEW

- Built using a React Native frontend paired with a lightweight Go backend to support scalability and modularity.
- Backend exposes RESTful API endpoints with proper error handling and structured JSON responses.
- Integrated two major external services:
 - OpenAI GPT-4o Mini for conversational cooking assistance
 - Spoonacular API for access to a recipe database of over 365k recipes
- Designed the system to maintain resilience across varying WiFi environments and mobile network conditions.
- Codebase organized into modular components to support future expansion into gestures and AR.

REACT NATIVE + EXPO

- Configured project with Expo for rapid iteration and access to mobile sensors.
- Implemented multi-screen navigation using React Navigation (Welcome → Search → Recipe → Cooking Steps).
- Developed a reusable UI system using a kitchen-friendly color palette (orange, gray, green).
- Patched dependencies and resolved security vulnerabilities discovered during development.
- Managed cross-platform quirks inherent to the Expo managed workflow.

VOICE INTERACTION & CORE FEATURES

- Voice interaction designed as the primary modality, but Expo restrictions required a simulated approach.
- Built voice-like input using text prompts and Alert.prompt to mimic command-based interaction.
- Supported hands-free-style features such as:
 - Ingredient-based search (natural language)
 - Step navigation (next, previous, repeat)
 - Smart timer interaction (start, pause, reset)
 - Clarification questions via GPT-4o Mini
- Added fallback logic for cases where recognition or network connectivity fails.

ADDITIONAL MODALITIES & FRAMEWORK

- Scaffolded a gesture framework for future mid-air interaction.
- Designed UI spacing and layout with future AR overlays in mind.
- Kept architecture modular so new input modes can be layered on without rewriting existing screens.
- Ensured long-term alignment with the project's Ubicomp goals.

AI & API INTEGRATIONS

- Integrated GPT-4o Mini for natural language cooking assistance.
- Built prompt structures that incorporate recipe steps and context for accurate responses.
- Implemented Spoonacular API for large-scale search, including multi-word ingredient queries.
- Added performance optimizations like two-stage loading (fast metadata → full data).
- Reduced unnecessary network calls and enhanced reliability across network types.

TECHNICAL CHALLENGES

- react-native-voice incompatible with Expo managed workflow.
- Microphone and camera access limited in development environment.
- Gesture recognition libraries require custom development builds.
- Physical device testing constrained by Expo's managed workflow.

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TESTING PLAN & NEXT STEPS

USABILITY TESTING GOALS

- Assess how intuitive the simulated voice-based navigation feels.
- Determine whether users can follow recipe steps with minimal device handling.
- Identify friction points in the search → recipe → cooking flow.
- Understand how well users trust and interpret AI-generated clarifications.

TESTING SCENARIOS

Participants will complete tasks such as:

- Searching for a recipe with natural-language ingredients.
- Navigating steps using simulated voice commands.
- Asking for step repeats or clarifications.
- Using the built-in timer (start/pause/reset).

These scenarios reflect realistic cooking behavior within MVP capabilities.

DATA COLLECTED

- Navigation ease and clarity of instructions.
- Points where users get confused or need help.
- Perceived cognitive load during cooking tasks.
- Quality and usefulness of AI clarifications.
- Overall satisfaction and feature expectations.

NEXT STEPS

Short-Term

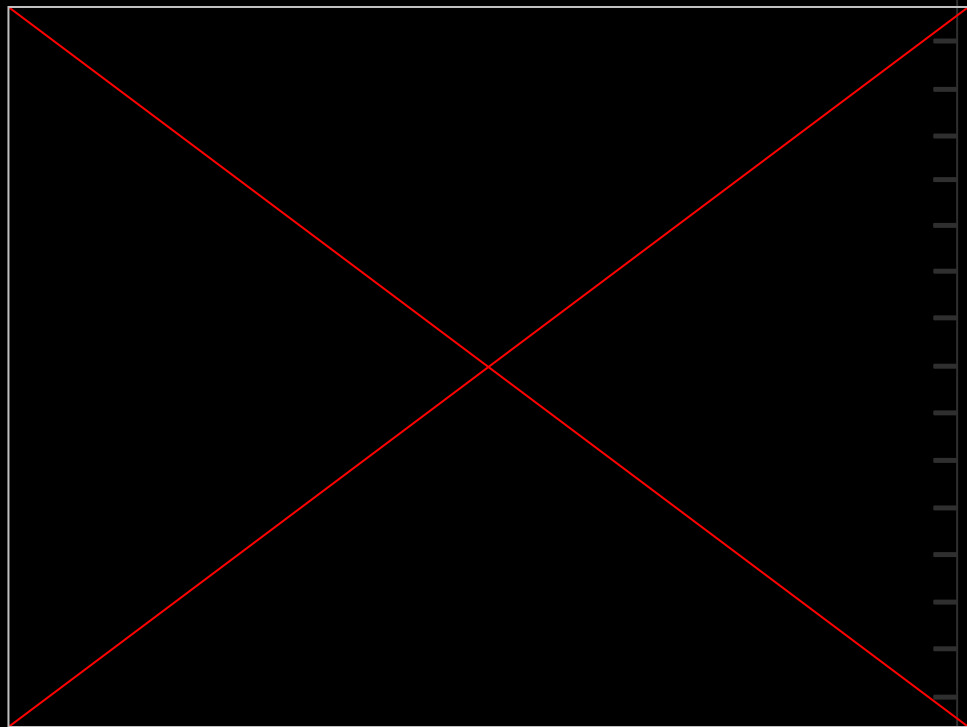
- Refine voice simulation and improve step transitions.
- Patch UI issues and enhance GPT prompt clarity.
- Strengthen fallback states and error messaging.

Medium-Term

- Implement real voice recognition through custom dev builds.
- Begin adding gesture input and early AR overlays.
- Improve performance and state management.

Long-Term

- Evolve NOMAD+ into a fully multimodal cooking assistant.
- Expand recipe intelligence and context awareness.
- Position the system as a platform for Ubicomp research.



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**THANK
YOU!**